

ARUN MEHRA

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Professional Summary

Software Developer with 9+ years of experience in Python, Go, Cloud Computing, Identity & Access Management (IAM), IoT Solutions, Cybersecurity and Generative AI (GenAI). Proven expertise in designing scalable microservices using AWS, GCP, and Azure. Skilled in Test-Driven Development (TDD), Object-Oriented Design (OOD), API Development, and Data Analytics. Experienced in Cloud SIEM, Threat Detection, DevOps Automation, LLM integration, LangChain, and AI-driven automation with a focus on security, compliance, and system optimization.

Core Skills & Technologies

- **Programming:** Python, Go, Linux Scripting
 - **Generative AI & ML:** LangChain, OpenAI API, Google Vertex AI, LLM Integration, AI-driven Automation
 - **Databases:** MySQL, PostgreSQL, MongoDB, Cassandra, SQLite
 - **Frameworks & Libraries:** Flask, Django, FastAPI, Pytest, Streamlit, Scrapy, Pandas, NumPy, OpenCV
 - **Cloud Platforms:** AWS (EC2, Lambda, CloudWatch), GCP, Azure
 - **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, Terraform, Prometheus, Grafana, Kafka, ELK Stack
 - **Security & IAM:** OAuth2, OpenID, JWT, RBAC
 - **Protocols:** REST, SOAP, MQTT, HTTP, IoT Matter Protocol
 - **Monitoring & Analytics:** AWS Kinesis, OpenTelemetry, Prometheus, New Relic, QuickSight
 - **Version Control & Agile Tools:** Git, Bitbucket, Jira, Confluence, Scrum
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Professional Experience

Lead Software Developer

Panasonic India Pvt. Ltd. | Oct 2021 – Aug 2025

Technologies: Python, Go, Flask, FastAPI, LangChain, OpenAI API, Google Cloud Vertex AI, Redis, Kafka, AWS Lambda, IoT Core, Mongo DB, Kafka, Kinesis,

- Spearheaded the integration of **Generative AI (LLM-based assistants)** into Panasonic's IoT ecosystem, enabling intelligent automation, natural language support, and personalized customer experiences.
- Integrated **LangChain with IoT device APIs** to provide contextual recommendations (energy saving, predictive maintenance, usage analytics).

- Led the development of **MirAle**, a cloud-based IoT platform integrating the **Matter Protocol** for cross-brand device interoperability.
 - Designed and deployed **scalable microservices architecture** for **device and home management**, ensuring high availability and reliability.
 - Developed **IAM solutions with Role-Based Access Control (RBAC)** to enhance security and multi-tenant access management.
 - Implemented **FOTA/LOTA updates** via **AWS Lambda and MQTT**, streamlining remote firmware upgrades and reducing operational overhead.
 - Led **DevOps & Security teams** to optimize **AWS cloud security policies**, improving incident response efficiency and system reliability.
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Software Engineer

FireEye Cyber Security | Oct 2020 – Oct 2021

Technologies: Python, Go, Celery, Django, Redis, PostgreSQL, AWS (Lambda, S3, Kinesis, CloudWatch, QuickSight), New Relic, Threat Analytics

- Developed **Cloud SIEM reporting tools** using **GraphQL & MongoDB**, automating security incident detection and response.
 - Designed and implemented an **AWS Cloud cost optimization tool**, leveraging **machine learning-based anomaly detection** to optimize resource usage.
 - Led collaboration with **DevOps & Security teams**, implementing **SIEM monitoring policies** to align with compliance standards and enhance security posture.
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Software Developer

Sierra Circuits | Jun 2019 – Sep 2020

Technologies: Python, KiCad, WxPython, Flask, FastAPI, Cassandra, AWS Lambda, AWS API Gateway, AWS S3

- Developed a **KiCad plugin** that automated PCB design validation, reducing design errors and improving accuracy.
 - Designed and deployed **Python-based microservices for PCB automation**, streamlining manufacturing workflows and reducing delays.
 - Created **Flask-based PCB analysis tools**, accelerating design verification and validation processes.
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Senior Technology Consultant

Aequm India Pvt. Ltd. | Jun 2016 – Jun 2019

Technologies: Python, Flask, FastAPI, MySQL, SQLite, Pandas, NumPy, OpenCV, Tesseract OCR, SOAP APIs

- Developed **Retail POS microservices**, enabling real-time data synchronization across multiple store locations.
 - Designed **AI-based invoice extraction models** using OpenCV and TensorFlow to automate invoice processing.
 - Implemented **CI/CD pipelines with Jenkins and AWS CodePipeline**, optimizing deployment workflows and reducing manual intervention.
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Key Projects

MirAle

Technologies: Python, Go, Flask, FastAPI, Redis, MQTT, Kafka, Kinesis, AWS (Lambda, EC2, CloudWatch, IoT Core), Google Cloud, React, LangChain, OpenAI API, Vertex AI

- **Engineered IoT microservices** for ACs, TVs, and third-party SDKs, achieving seamless cross-brand interoperability via the Matter Protocol.
 - **Integrated Generative AI assistant** (LLM-powered with LangChain & Vertex AI) for natural language device interaction, contextual recommendations, and predictive maintenance insights.
 - **Built a Virtual Device Service** for simulation and testing, accelerating development cycles and reducing troubleshooting effort.
 - **Developed Energy Management & Optimization solutions**, combining IoT telemetry with GenAI-driven analytics to provide personalized energy-saving recommendations.
 - **Enabled voice + text interaction** with Amazon Alexa, Google Home, and GenAI assistants, enhancing customer engagement and support.
 - **Implemented secure RESTful APIs** supporting real-time device communication over MQTT and HTTP, with multi-tenant RBAC for scalable IAM.
 - **Optimized event-driven architecture** using Kafka & AWS Kinesis for high-throughput data streaming, AI-driven anomaly detection, and real-time monitoring.
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Helix XDR (Cloud SIEM Solution)

Technologies: Python, Go, Celery, Django, Redis, PostgreSQL, AWS (Lambda, S3, Kinesis, CloudWatch, QuickSight), New Relic, Threat Analytics

- Developed **secure APIs for UEBA (User and Entity Behavior Analytics) tools**, improving **threat detection capabilities** and **automated incident response**.
- Built **real-time monitoring dashboards** using **New Relic and AWS QuickSight**, enhancing **system observability, performance monitoring, and security insights**.
- Designed and implemented **Threat Analytics pipelines** leveraging **machine learning models** to detect **anomalous activities, intrusion attempts, and insider threats**.
- Integrated **AWS Kinesis and CloudWatch** for **real-time log streaming and security event processing**, improving **incident detection speed and response times**.
- Optimized **data ingestion and correlation engines**, reducing **false positives in threat detection** and improving **SIEM analytics efficiency**.

- Collaborated with **security and DevOps teams** to refine **SIEM monitoring policies**, ensuring compliance with **industry standards and security best practices**.
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Sierra Companion App

Technologies: Python, KiCad, WxPython, Flask, FastAPI, Cassandra, AWS Lambda, AWS API Gateway, AWS S3

- Developed a **real-time PCB design validation plugin** using **KiCad and WxPython**, improving **design accuracy and reducing human errors**.
 - Designed and implemented **microservices-based architecture** for **PCB design automation**, ensuring **scalability and modularity**.
 - Built **serverless APIs using AWS Lambda & API Gateway**, enabling **on-demand processing** while reducing infrastructure costs.
 - Integrated **PCB design libraries** with **Python-based automation scripts**, streamlining **design verification and validation processes**.
 - Utilized **AWS S3 and Cassandra** for **storing and retrieving design schematics**, ensuring **efficient data access and version control**.
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99POS System (Retail POS Solution)

Technologies: Python, Flask, FastAPI, MySQL, SQLite, Pandas, NumPy, OpenCV, Tesseract OCR, SOAP APIs

- Developed a **comprehensive Point of Sale (POS) system** with **billing, inventory, and sales management features**, improving **retail operations and transaction efficiency**.
 - Designed and implemented **RESTful APIs using Flask and FastAPI** to facilitate **seamless interaction between the POS system, inventory database, and external services**.
 - Integrated **OCR-based document scanning using OpenCV & Tesseract**, automating **invoice and receipt processing**, reducing **manual data entry errors**.
 - Built a **robust inventory tracking system** with **real-time stock updates, order processing, and sales forecasting** using **MySQL and SQLite**.
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Education

- **PG Diploma in Embedded System Design** – CDAC Bangalore (Aug 2015)
- **B.E. in Electronics & Communication Engineering** – Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal (June 2014)